Journal of Sensory Studies

WILEY

Journal of Sensory Studies

ORIGINAL ARTICLE

Co-Creation of Complementary Food for Operational Rations: An Exploratory and Collaborative Case Study Between Brazilian Military Personnel

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Received: 18 September 2024 | **Revised:** 29 November 2024 | **Accepted:** 30 January 2025

Funding: This work was supported by National Council for Scientific and Technological Development (CNPq) and Carlos Chagas Filho Research Support Foundation of the State of Rio de Janeiro (FAPERJ) for the grant provided currently to the first author (Grant 150908/2023-7).

Keywords: co-creation ration | military acceptance | military foods | nutritive bar | operational

ABSTRACT

In the development of operational rations, products designed to meet military demands must support adequate nutritional intake to enhance performance in missions and operations. In this sense, the present study adopted a co-creation methodology to develop a product intended for inclusion in operational rations, guided by the perceptions of Brazilian military personnel with experience in using these rations. An online questionnaire was conducted to collect both structured and unstructured data, framed as a co-creation task involving 360 members of the Brazilian Army. The results revealed a clear preference for a product with a high protein content (specifically, whey protein), chocolate flavor, bar format, and soft texture among different presentations, flavors, and formulations. This approach addresses existing questions about developing optimal products for military personnel and offers industrial and academic insights for tailored product development in specialized sectors.

1 | Introduction

Operational ration packs are the sole source of nutrition when military personnel cannot access fresh food and field kitchens due to deployment and training in remote and hostile locations. They should be low-weight, durable, nutrient-rich, and contain sufficient energy to ensure personnel can perform their duties (Lenferna De La Motte et al. 2023). Kullen et al. (2016a) described the following as relevant criteria for feeding military personnel: diet quality, the choice of healthy food, and fulfilling specific nutritional requirements. Diet quality involves nutritional composition influenced by raw materials, other ingredients, preparation, and storage, affecting operational rations' acceptability.

The basic premise is that operational rations must provide nutrients in a balanced manner and in varied situations to support combatants in their designated missions. Thus, the Brazilian Army's combat operational ration was designed to sustain a soldier in operations for a 6, 12, or 24-h period, consisting of a set of main basic foods (thermoprocessed meals in retort pouch packaging, in this case, lunch or dinner), complementary food items (cassava flour, coffee, chocolate milk, sugar, hydro electrolyte replenisher, and protein bar, among

Summary

- The findings of this study have direct implications for the development of operational rations tailored to the preferences and needs of military personnel.
- By adopting a co-creation methodology, the research provides valuable insights into the specific characteristics of complementary foods that enhance acceptance and consumption in military contexts, particularly a high-protein, chocolate-flavored bar with a soft texture.
- These results can guide the food industry in developing products that align with soldiers' nutritional and operational demands, contributing to improved mission performance.
- Additionally, the study highlights the utility of co-creation methods for product innovation in specialized sectors.

others), and accessories for cooking, such as stoves and cutlery, which are provided when it is impossible to deploy a field kitchen, and should have good acceptability by the supported troop (Brasil 2022).

None of the components present a unique employment profile for military personnel; that is, they were not subjected to any development processes that meet the specific demands of combatants despite unrelated consumption conditions. The technical specifications of Brazilian operational rations are defined in Ordinance No. 1.416 of the Ministry of Defense from 2008, inferring opportunities for developing new products in their cataloging (Brasil 2008).

Therefore, despite technological innovations in nutrition, health, and performance, the food products offered remain largely unchanged over time, negatively impacting the consumption of these items. According to a study by (De Abreu et al. 2023), there is a high incidence of monotony, low acceptability, and underconsumption in various menus. This scenario directly affects the health and performance of combatants, underlining the urgent need to develop nutritionally enriched and balanced foods that better meet the specific requirements of military operations (De Abreu et al. 2024).

Consequently, military personnel should be closely involved in new product development to ensure that healthy alternatives meet their needs. It is essential to highlight the context of their activities, which must also be considered when developing new products: military personnel are subjected to hostile environments, including high-altitude terrain, the tropics, deserts, and sub-zero temperatures; this leads to soldiers suffering many types of physical and psychological stress. Proper nutrition is thus highly relevant to mitigating weight loss and boosting physical, cognitive, and immunological performance (Tassone and Baker 2017). Scientific knowledge has reached a point where creating healthier and operationally effective products is plausible and imperative. It involves a technical integration of performance, health, operability, nutrient enrichment, and considerations of presentation, flavor, weight, volume, packaging, and consumption convenience (De Abreu et al. 2023).

In this context, co-participatory techniques, such as the co-creation methodology, should be used for innovation and product development. This approach in new product development is defined as "a collaborative new product development activity in which consumers dynamically contribute and select various elements of a new product offering," and emerges as a salient form of consumer participation (Mandolfo et al. 2020). Active involvement allows participants to reflect creatively on their practices, making it a promising approach to tackling public health issues, where the reassessment of current practices is necessary and top-down solutions fall short (Leask et al. 2019).

Researchers in the military context have recently used the co-creation methodology. (Liwång 2022), Studies on policy and technology to develop concepts for future capability systems for defense and security identified that co-creation is a specific form of research collaboration that facilitates results that would not have been possible without a joint approach. Andersson (2018), exploring development projects, a combat vehicle, and a naval vessel, described the role of demonstrators and the close cooperation between research, government agencies, and industry, using co-creation, identifying that this methodology promotes fruitful dialogue on central capability aspects and challenges.

It should be noted that there is still no methodological definition of the tool, and different ways of executing co-creation have already been successfully used, such as through focus groups, gamification, crowdsourcing, and online co-creation, which companies and similar entities are already using for the development and commercialization of new products (Banović et al. 2016; Bharati et al. 2021; Rodrigues 2019). In this context, an online co-creation technique would be a suitable way to obtain, with quality and precision, information and impressions in developing new products directly from end-users (military personnel from the Brazilian Army), regardless of the geographic location of the military work base.

In this sense, the present work aimed to adopt an online and collaborative approach to identify essential characteristics deemed necessary by the military to be incorporated into a new nutritionally enriched product, designing a nutritional supplement aimed at enhancing both military operational efficiency and diet quality, aligned with the parameters and needs elicited and required observed by De Abreu et al. (2024). It is expected that the findings of this research will lay the foundations for developing an innovative product prototype intended for implementation in the Brazilian Army. This strategic effort aims to promote a comprehensive review of Brazil's current military food cataloging regulations, ensuring their alignment with emerging nutrition and operational performance needs. Furthermore, it is expected that the methodological approach adopted and the knowledge obtained will contribute significantly to future studies in military nutrition and the development of innovative solutions for the tactical food industry.

Operational ration packs are the sole source of nutrition when military personnel cannot access fresh food and field kitchens, usually during deployment and training involving remote and hostile locations. Rations should generally be lightweight, durable, nutrient-rich, and provide sufficient energy to perform expected duties. Kullen et al. (2016b) posited that diet quality, the choice of healthy foods, and meeting specific nutritional requirements are priority criteria in military rations. Diet quality involves nutritional composition, which is influenced by the choice and quality of raw materials and the preparation and storage processes.

The basic premise is that operational rations must provide nutrients in a balanced manner and in varied situations to support combatants in their designated missions. The Brazilian Army's combat operational ration was designed to sustain a soldier in operations for 6, 12, or 24 h. In general, it consists of two main dishes offered in the form of thermoprocessed meals in retort packaging, complementary items (cassava flour, coffee, chocolate milk, sugar, hydroelectrolyte replenisher, protein bar, among others), and cooking accessories, such as stoves and cutlery, which are provided when it is impossible to deploy a field kitchen.

Although widely used by Brazilian military personnel, none of the ration items present a unique employment profile for military personnel, meaning they have not undergone development processes to meet the specific demands of combatants. More specifically, the technical specifications of Brazilian operational rations were defined in Ordinance No. 1.416 of the Ministry of Defense in 2008 and have had their criteria fixed since then (Brasil 2008). Thus, despite the technological advances observed in the last two decades in nutrition, health, performance, and food technology, the food products included in Brazilian military operational rations remain largely unchanged over this period, which has caused negative impacts on the consumption of these items. According to a study by De Abreu et al. (2023), there is a high incidence of monotony, low acceptability, and under-consumption in various menus included in the rations, and it is common for individuals to carry personal items as supplements to their rations, items available in conventional markets and not customized for operational rations.

Inadequate nutrition and low hedonic response to meals are directly related to the health and performance of combatants (Carins et al. 2022), underlining the urgent need to develop new foods that are not only nutritionally enriched and balanced but also better meet the specific requirements of military operations. One way to optimize the development or reformulation of foods, especially those with well-defined consumer niches, involves involving the target audience in the process (Beacom et al. 2021); in the case of operational rations, this consists of considering the opinions of the military who consume the rations, correlating them with the context of the activities they perform. Given that military personnel are subject to hostile environments that include high-altitude terrain, tropics, deserts, and sub-zero temperatures, leading them to suffer many types of physical and psychological stress, these factors must be included in the development process. Therefore, the objective of this study was to co-create the concept of a supplementary item for operational rations from the perception of Brazilian military personnel.

2 | Materials and Methods

2.1 | Participants and Sampling

Three hundred sixty military personnel ($n = 360$) were recruited by disseminating an invitation poster on the Brazilian Army's official communication channels. In addition to the transmitted invitation, participants were asked to share the research with other military personnel in different parts of the country, describing the sampling process as a mix of convenience and snowball (Raifman et al. 2022). The inclusion criteria were military labor status and previous contact with operational rations.

In the co-creation methodology, the minimum sample size is not rigidly defined or based on statistical calculations typical of probabilistic sampling, as co-creation has a qualitative background focused on the depth and richness of the data collected, aiming more precisely at the saturation of ideas or concepts rather than making population inferences (Creswell and Poth 2018). For example, co-creation has been applied in various contexts with different numbers of participants: a study on hybrid meat products in Europe involved 2405 people, divided among Denmark, Spain, and the United Kingdom (Ryder et al. 2023). Other studies included 264 Finnish military personnel to identify desirable characteristics in troop transportation (Juntunen 2012) and 52 Uruguayan children in a project on healthy foods for school programs (Velázquez et al. 2022). This study was approved by the Ethics and Research Committee of the Federal University of Amazonas, under-enrollment CE53496121.1.0000.5020.

2.2 | Online Survey

An online questionnaire was structured on the Compusense Cloud platform and completed by 360 members of the Brazilian Army between June and August 2023. Before distributing the online questionnaire, it was administered to 20 military personnel to assess their understanding of the data collection instrument. This preliminary application served as a critical step to ensure the clarity and effectiveness of the questionnaire in capturing the intended information. Based on the feedback and observations gathered from this initial group of participants, necessary adjustments were made, improving the overall coherence of the investigation and the understanding of the target audience, thereby allowing the questionnaire to reflect the research objectives accurately.

2.2.1 | First Part: Sociodemographic Questions

The study initially focused on collecting sociodemographic data, including age, marital status, education, family income, military rank, and geographic location. This preliminary information serves as a foundation for understanding military personnel's diverse lifestyles and requirements.

2.2.2 | Second Part: The Need to Develop a Supplementary Product for Operational Rations and Underlying Related Reasons

The appropriateness of designing a nutritionally enriched supplementary item for operational rations in alignment with the reality of military service was evaluated through a Likert scale question rated on a 7-point scale, ranging from “Absolutely Inappropriate” to “Absolutely Appropriate,” accompanied by an open-ended question that sought reasons related to the level of appropriateness of the research with the reality of Brazilian troops (Table 1). This section encouraged participants to express opinions, share perceptions about their experiences with existing products, and

TABLE 1 | —Assessment of the project's suitability and the product's characteristics to be developed.

Considering the creation of a new “nutritionally enriched product” to accompany the operational ration, please indicate on the scale below how appropriate you consider this act/project	1—Absolutely inappropriate 2—Absolutely appropriate 3—Inappropriate 4—Slightly inappropriate 4—Neutral 5—Slightly appropriate 6—Appropriate 7—Absolutely appropriate
Open-ended question	In brief, please explain the reasons for your choice above
Considering the development of an enriched product for operational rations, please indicate the importance of each characteristic for you:	
Be healthy	1. Not important at all 2. Slightly important 3. Important 4. Very important 5. Extremely important
Be nutritionally balanced	
Have a low environmental impact	
Not generate a large volume	
Not significantly contribute to the weight to be carried	
Increase satiety	
Be palatable (pleasant to the taste)	
Be a safe food	
Have good digestibility	

elaborate on the perceived relevance of the project within the field. The procedures followed Grasso et al. (2022), with modifications.

2.3 | Third Part: Desired Attributes

The key attributes for military rations, previously identified by De Abreu et al. (2024), were evaluated in this research using a five-point Likert scale, ranging from 1 (not important at all) to 5 (extremely important). In this phase, participants expressed how important they considered each attribute when developing an ideal product in operational rations. The key attributes addressed included various aspects such as health, nutritional balance, product volume/size, transported product weight, satiety, palatability, safety, and digestibility.

2.3.1 | Fourth Part: Product Formulation

In the final part of the questionnaire, participants ranked various intrinsic and extrinsic attributes related to the product to be developed. This process included five main sections, by studies by De Abreu et al. (2024), evaluated as shown in Table 1.

2.4 | Data Analysis

The sociodemographic data collected, the question related to the project's alignment with the reality of the troops, and the categories that emerged from the open-ended question were evaluated according to their observed frequencies for the studied sample. Specifically, for the open-ended question, a content analysis (Bardin 2011) was structured through a thematic framework (Mesías et al. 2021), where the information was organized into common themes. An inductive method was employed for the analysis, allowing categories to emerge organically without predefined categories based on the dimensions of the questionnaire as initial coding (Kyngäs 2020). The elicited phrases and words were then coded, with an initial search for recurring terms within each stimulus. Terms with similar meanings were grouped into categories and themes that semantically encompassed them. Three independent researchers analyzed the data, and after individual analyses, they used the principle of author triangulation to compare the results (Restivo et al. 2023). This method aims to achieve consensus by contrasting different perspectives and interpretations. The most frequently mentioned terms were selected, and similar terms were consolidated into the same category. Categories were considered significant for the analysis when mentioned by more than 5% of participants, a cutoff point that, according to previous studies, helped avoid the loss of a significant amount of information (Sass et al. 2018).

The analysis of the importance attributed to the attributes considered strategic in the product to be developed, in the context of operational rations, present in the third block of the questionnaire, was conducted considering the Likert scale data as continuous after checking normality via Shapiro–Wilk. Therefore, an analysis of variance with Tukey's HSD post hoc test ($p < 0.05$) was conducted.

The Friedman test was used to rank characteristics intended for incorporation into the prototype model (part IV of the questionnaire), given the non-parametric nature of the data from the ranking task (Pereira et al. 2015). For a more robust comparative analysis of the parameters under study, the Nemenyi post hoc statistical test was chosen. This non-parametric method is particularly suitable for comparing multiple groups and is used here to evaluate differences between the scores attributed to various parameters. It allows a detailed investigation into significant disparities in the recorded preferences, maintaining statistical rigor even in non-normal data distribution (Agans et al. 2020).

All the statistical analyses were performed using the SPSS Software (version 26, IBM Inc., Chicago, IL, USA) and XLSTAT 2022.4.5 Software (Addinsoft, Paris, France).

3 | Results

3.1 | Sociodemographics

Table 2 presents the sociodemographic data obtained for the sample. The most represented age group is 18–25 years, comprising 30.26% of the participants, indicating a predominance of young adults, consistent with other studies involving military samples (Reyes-Guzman et al. 2015; Goodwin et al. 2015). The age group of 36–45 years constituted 25.8%

of the sample, and those over 45 years accounted for 27.4%, the latter being uncommon in the literature involving military personnel. The age group of 26–35 years had the lowest representation at 16.2%.

Regarding marital status, 36.3% of the participants were single, 30.26% were married, and 14.4% were in a stable union. Lower frequencies were observed for divorced (3.7%) and widowed (0.4%) individuals. Regarding education, 45.3% of the respondents had postgraduate degrees, suggesting a high level of educational attainment. High school and undergraduate degrees also had notable representations, with 19.5% and 17.3%, respectively.

Economically, 36.1% of the military personnel had incomes above 10 times the minimum wage, which may be associated with their high level of education. Additionally, 30.2% of the military personnel held ranks of captain or higher positions, indicating a significant representation of leadership roles within the sample.

Furthermore, those classified as "Captain/Lieutenant/Asp" (30.2%) and "Major/Lieutenant Colonel/Colonel" (30.2%) had the highest frequencies, suggesting a significant representation of professionals in leadership or higher positions. Most participants resided in the southeastern region of the country (56.5%), in contrast to the northern region, which accounted for only 5.3%.

3.2 | Need for Developing a Complementary Product for Operational Rations and Related Underlying Reasons

The average score observed regarding the suitability of developing a new food product for operational rations was 6.28 on a scale of 7. Therefore, most respondents indicated that it was Appropriate or Absolutely Appropriate. This positive trend strongly indicates a demand or a significant openness to innovations or improvements in current products. The results observed in the open question provided a comprehensive view of the various dimensions that military personnel may consider when adding a product to their operational rations. The data collected from the open-ended question were subjected to content analysis and organized into categories that grouped terms with semantic equivalence, finally using a thematic framework to manage the categories into underlying themes (Table 3). Four main themes were identified, with 12 categories mentioned above the cutoff limit of 5% and 8 categories below the cutoff limit. Table 3 presents the identified themes and the categories they comprise, along with examples of responses that belong to the respective categories.

In total, responses from 298 of the 360 military personnel included in the study utilized open-ended questions to present the reasons related to how appropriate they found the research involving the development of a complementary item for operational rations. The remaining participants provided responses such as "nothing to declare" and "prefer not to answer." The 298 responses yielded 595 classifications into underlying categories,

TABLE 2 | —Age, marital status, education level, house incoming, graduation, and region of the military.

Attribute	Item	Percentage
Age	18 and 25 years old	30.60%
	26 and 35 years old	16.20%
	36 and 45 years old	25.80%
	Over 45 years old	27.40%
Marital status	Single	36.30%
	Stable union	14.70%
	Married	44.90%
	Divorced	3.70%
	Widower(s)	0.40%
Schooling	No schooling	0.40%
	Incomplete elementary school	0.40%
	Complete elementary school	0.70%
	Complete high school	19.50%
	Technical education	1.50%
	Incomplete higher education	14.20%
	Complete higher education	17.30%
	Postgraduate studies	45.30%
	I prefer not to inform	0.00%
Household income	Up to 1 minimum wage (up to R$ 1320.00)	6.30%
	From 2 to 5 minimum wages (R$2640.00 to R$6600.00)	23.00%
	From 6 to 10 minimum wages (from R$ 7920.00 to R$ 13,200.00)	29.10%
	Above 10 minimum wages (R$13,200.00)	36.10%
	I prefer not to inform	5.50%
Rank/graduation	Student	8.30%

(Continues)

TABLE 2 | (Continued)

Attribute	Item	Percentage
	CPL/PVT	17.90%
	SGT/WO	13.30%
	Captain/lituant	30.20%
	Major/Lituant Colonel/Colonel	30.20%
Region	Northeast region	20.10%
	North region	5.30%
	Midwest region	12.00%
	Southeast region	56.50%
	South region	6.10%

as many responses were related to more than one category due to presenting reasons for different semantic values.

The theme "Nutrition and health of military personnel" was the most cited, with 218 mentions (73.15%). Overall, for this theme, the responses highlighted the need to provide healthy, nutritious, and safe foods to meet the intense energy demands of military work, emphasizing the importance of adequate nutritional value to improve operational efficiency and soldiers' endurance in combat. Within this theme, the category "Promote adequate nutrition" was the most frequent, with 143 mentions (47.99%). The high representativeness of this category suggests that Brazilian military personnel pay significant attention to the nutritional value of the food they consume during operations. Similarly, the analysis of a large portion of the other categories indicates utilitarian values among the military, that is, what they expect a nutritionally adequate product to help them achieve. The category "Meet high energy demand" was also significant, with 30 references (10.07%), underscoring the need for foods that support the high caloric expenditure of military activities. Additionally, "Promote health" (24 mentions, 8.05%) and "Promote a healthy diet" (21 mentions, 7.05%) were highlighted as reasons why developing a complementary item for rations is appropriate, indicating that military personnel believe a nutritionally enriched product would positively impact their health.

The second most cited theme was "Physical and operational performance," with 180 references (60.40%). The category "Ensure operational capability" was the most prominent, with 98 mentions (32.89%), emphasizing the relevance that military personnel place on operational capability and the relationship between nutrition and the ability to maintain physical and mental fitness to meet all technical requirements of operations, with operational capability being decisive for preserving physical integrity in risk situations. Similarly, the category "Promote performance enhancement (ergogenic effect)" (38 mentions, 12.75%) highlighted the importance of any improvement that could enhance military performance. Furthermore, "Sustain intense routine" was mentioned in 28 responses (9.40%), underlining the extreme physical demands of military activities and the need to perform them effectively.

The theme "Perceptions regarding updates" was cited in 71 responses (23.83%). The category "Correct deficiencies in the current ration" was the most frequent, with 25 mentions (8.39%), pointing out deficiencies in the current rations, such as the lack of consistent protein and the insufficiency of some meals. "Ideas for continuous improvement" (20 mentions, 6.71%) and "Positive judgment regarding the development of a new product" (26 mentions, 8.72%) were also mentioned, expressing that some military personnel believe that rations should be subject to constant improvements and incorporation of new technologies.

In the theme "Sensory aspects and consumption-related factors," 40 mentions were identified (13.42%). The category "Improve palatability" was cited 24 times (8.05%), highlighting the need for a diet that is not only nutritious but also has a pleasant taste. Within this topic, rejection and waste due to low sensory appeal were observed. "Increase practicality" (16 mentions, 5.37%) emphasized the need for practical products that are easy to consume and meet nutritional needs with minimal time expenditure.

Finally, the categories with lower representativity, those that did not reach the 5% cutoff but still may carry relevant information, totaled 96 mentions (32.21%). Among these, "Utilize the experience of the military in development" (13 mentions, 4.36%) was considered significant when developing a new product to be added to the ration, as well as developing such a product could help "Reduce monotony" (9 mentions, 3.02%), "Reduce carried load/Improve logistics" (12 mentions, 4.03%), "Minimize adverse effects of consumption" (10 mentions, 3.36%), "Boost morale" (11 mentions, 3.69%), "Promote innovation in the armed forces" (10 mentions, 3.36%), "Adapt rations to regional preferences" (10 mentions, 3.36%), and "Increase satiety" (10 mentions, 3.36%). These categories provide valuable insights into the importance of considering the experience of military personnel, reducing food monotony, improving logistics, minimizing adverse consumption effects, boosting morale, innovating in the armed forces, adapting rations to regional preferences, and increasing satiety.

3.3 | Desired Attributes

The analysis of participants' priorities and preferences regarding intrinsic and extrinsic attributes considered crucial in developing a complementary product for military rations (Table 4) indicated higher averages for Safety (4.6) and Digestibility (4.5), with no significant difference ($p < 0.05$) between them.

The attributes palatability (4.4), satiety (4.4), low volume (4.4), nutritionally balanced (4.3), low weight (4.3), and healthy (4.3) showed similar averages, not differing from Digestibility, which was in the same group as Safety. The attribute of low environmental impact was the least valued (3.6 ± 1.16, group d), although still considered relevant within the overall context. The low environmental impact attribute was the least valued (3.6) among the military personnel.

TABLE 3 | —Synthesis of themes, categories, and responses on the relevance of a research project for military operational rations.

Theme	Categories/examples
Nutrition and health of the military personnel ($n=218/73.15\%$)	Promote adequate nutrition ($n=143/47.99\%$)
	"The troops require healthy, nutritious, and safe foods to meet the intense energy demands of their work." "It is important to have food that adds nutritional value and enhances operational efficiency." "The operational ration must be designed to meet the nutritional needs of soldiers in combat, consequently providing nutrients to improve their endurance"
	Promote health ($n=24/8.05\%$)
	"A nutritionally enriched product will positively impact the combatant's health, increasing the troops' operational effectiveness." "Due to the need to maintain the immune system through essential vitamins and supplements for health." "More health and energy to be utilized in combat"
	Promote healthy diet ($n=21/7.05\%$)
	"Healthy eating is important for maintaining operational capacity." "It contributes to a balanced diet, ensuring that physiological functions are performed adequately." "A product that provides a healthier meal"
	Meet high energy demand ($n=30/0.07\%$)
	"The troops require healthy, nutritious, and safe foods to meet the intense energy demands of their work." "Military personnel in operations tend to have higher caloric expenditure, making it plausible that a nutritionally enriched product would support their health." "Due to the high energy consumption of military activities, it is crucial to have a product that complements the operational ration"
Physical and operational performance ($n=180/60.40\%$)	Ensure operational capability ($n=98/32.89\%$)
	"The relevance of developing a nutritious product that meets operational needs." "It is very important for maintaining combat effectiveness." "The military personnel need a diet capable of sustaining them during field operations"
	Promote performance enhancement (ergogenic effect) ($n=38/12.75\%$)
	"Anything that can enhance the performance of military personnel is welcome." "Improving the performance of military personnel." "For military personnel in the field, the more nutritious the food, the better their performance will be"
	Sustain intense routine ($n=28/9.40\%$)
	"Military activity is extremely demanding. Therefore, I believe there is a need for dietary supplementation." "Operational rations are meant to be used when military personnel are typically engaged in strenuous physical activities." "In operational activities, where physical exertion is high, it is very important for military personnel to maintain adequate nutrition levels in order to perform their mission well"
Perceptions regarding updates ($n=71/23.83\%$)	Idea of continuous improvement ($n=20/6.71\%$)
	"Although the current operational ration is good, any enhancement is always welcome". "Adequate support for completing a mission must be as suitable as possible and continually improved". "I believe improvements are necessary"
	Correct deficiencies in the current ration ($n=25/8.39\%$)
	"Enhancing our operational ration is very important, considering the current situation of outdated rations." "The coffee and the dinner of the ration are extremely deficient, and a consistent protein is also missing in some meals." "Although the R2 ration is sufficient for lunch, dinner, and supper, it does not meet the needs for breakfast"
	Positive judgment regarding the development of a new product ($n=26/8.72\%$)
	"It is of utmost importance to always think about improvement." "This study is very important." "The study is highly necessary"

(Continues)

Theme	Categories/examples
Sensory aspects and consumption-related factors ($n = 40/13.42\%$)	Increase practicality ($n = 16/5.37\%$)
	"The operational ration needs products that are easy to consume and meet the nutritional needs of the combatant with minimal time expenditure. Military personnel do not consume many products in the operational ration." "The development of a new product must be palatable and easy to prepare." "It should be a product that is easier to handle and has an acceptable taste"
	Improve palatability ($n = 24/8.05\%$)
	"The current ration has a terrible taste." "Military operations cause significant wear and tear on participants and demand food rich in nutrients and pleasant taste." "To maintain a nutritious and tasty diet"
Categories with lower representativity ($n = 96/32.21\%$)	Utilize the experience of the military in development ($n = 13/4.36\%$)
	"Opportunity for the end user to express their opinion on the product they will consume"
	Reduce monotony ($n = 9/3.02\%$)
	"The current quantity of the ration is insufficient, and the ration is repetitive"
	Reducing carried load/improving logistics ($n = 12/4.03\%$)
	"The product should be tasty, rich in carbohydrates or protein, and small and lightweight. A negative example of this is the rice in the R-2 ration, as it is unappetizing, heavy, and bulky. Moreover, some military personnel prefer to discard this food rather than consume it"
	Minimize adverse effects of consumption ($n = 10/3.36\%$)
	"The current ration is high in calories and causes various adverse effects, such as gas, nausea, and others"
	Boost morale ($n = 11/3.69\%$)
	"Nutrition is essential for maintaining the morale of the combatant, both in training, in combat, and throughout their career"
	Promote innovation in the armed forces ($n = 10/3.36\%$)
	"Updating with market practices and innovations is necessary"
	Adapt rations to regional preferences ($n = 10/3.36\%$)
	"I believe this initiative will contribute to having a high-quality operational ration that adapts to various climates and does not deteriorate during military exercises such as marches and field exercises"
	Increase satiety ($n = 10/3.36\%$)
	"Military personnel need a diet capable of sustaining them during field operations"

3.4 | Product Formulation

Table 5 provides a detailed presentation of the results obtained in prioritizing the expected characteristics of the complementary product for the operational ration according to the preferences of the military personnel.

3.5 | Presentation

The "Powder" formulation was the least preferred (mean 5.34) and showed significant differences among the tested options ($p < 0.05$). "Cakes," well-appreciated products in conventional consumption settings, seem to be less desirable in operational situations, being the second least preferred presentation (mean 4.35) and statistically similar only to squeeze gel ($p < 0.05$). "Gel Squeeze" showed low preference among combatants (mean 4.22).

The presentation of snacks (processed foods fried in oil, typically made from dough or potatoes, known for their crunchy texture and salty flavor) showed statistical similarity to biscuits/cookies and mixed grains/cereals with a higher average (3.82). The presentations of "Bars" (mean 3.25), "Mixed Grains and Cereals" (mean 3.51), and "Biscuits/Cookies" (mean 3.51) showed no significant statistical differences ($p < 0.05$), making them the most preferred options by military personnel. The similarity in average ranking indicates that from a nutritional and convenience standpoint, mixed grains/cereals and biscuits/cookies are comparable to bars.

TABLE 4 | —Comparative assessment of values for desired attributes in the development of food products for the military.

Attribute	Values*
Safety	4.6 ± 0.66 [a]
Digestibility	4.5 ± 0.69 [a,b]
Palatability	4.4 ± 0.78 [b,c]
Satiety	4.4 ± 0.78 [b,c]
Low volume	4.4 ± 0.78 [b,c]
Nutritionally balanced	4.3 ± 0.79 [b,c]
Low weight	4.3 ± 0.89 [b,c]
Healthy	4.3 ± 0.93 [c]
Low environmental impact	3.6 ± 1.16 [d]

*Values are expressed as mean ± standard deviation ($n = 360$). Different letters in the same column indicate significant differences between samples ($p < 0.05$) by Tukey's test.

3.6 | Texture

According to Table 5, the "Soft" texture is the most desired (2.12; $p < 0.05$), differing significantly from "Crunchy" (2.52; $p = 0.014$), "Fibrous" (3.23; $p < 0.0001$), "Pasty" (3.40; $p < 0.0001$), and "Liquid" ($p < 0.0001$), reinforcing its preference. "Liquid" products scored the highest average, indicating the least preference among the military personnel. "Pasty" formulations ranked as the second least preferred option (mean 3.40). The "Fibrous" presentation did not differ from "Pasty."

The second preference in the ranking, "Crunchy," shows significant differences compared to other presentations ($p < 0.05$). Finally, and foremost in the study's preferences, the "Soft" texture was statistically distinct from all others ($p < 0.05$).

3.7 | Flavor

According to Table 5, the "Other Fruits" flavor was significantly less preferred (5.80; $p < 0.05$) by the military personnel, highlighting a tendency to favor familiar and widely accepted flavors. The flavors "Vanilla" and "Cookies and Cream," with a p-value of 1.000, show no significant difference in preference.

"Acai," "Dulce de Leche," and "Citrus Fruits" have similar mean preferences and are only statistically different from chocolate and other fruits ($p < 0.05$). "Strawberry," ranked as the second preferred option (mean 4.06), is statistically similar to most other flavor formulations. Statistically distinct from all other flavors, "Chocolate" was selected as the preferred flavor (mean 3.28; $p < 0.05$).

3.8 | Nutritional Enrichment

Despite being considered valuable for skin and joint health, Collagen does not appear to be preferred by military personnel, presenting a mean value of 6.97 ($p < 0.05$). "Caffeine," "Maltodextrin," and "Omega-3" also show low preference rankings with no significant statistical differences ($p < 0.05$). "Fibers," "Minerals," and "Vitamins" did not show significant statistical differences among them and received intermediate rankings, suggesting a consistent appreciation for their health benefits ($p < 0.05$). "Complex Carbohydrates" share a slightly lower (3.53) mean than the listed items above, presenting a significant difference. "Protein" has significantly the lowest average (2.24; $p < 0.0001$), indicating that it is the most preferred item with relatively low variability in responses compared to the nutrient sources investigated.

3.9 | Protein Source

According to Table 5, the most preferred protein source among the soldiers was "Whey Protein" (1.96; $p < 0.0001$). Then, the preferred protein sources were "Albumin" (2.58) and "Casein" (2.92), with no statistically significant differences between them ($p = 0.062$). "Plant-based" proteins ranked moderately as the fourth choice for development (2.90), aligning with the growing demand for healthier products.

4 | Discussion

The observed age values (majority 18–25 years) indicate a predominance of young adults, aligning with other studies involving military samples (Goodwin et al. 2015; Reyes-Guzman et al. 2015). Additionally, the high level of education and income observed in the sample corroborates findings by Ferreira and Torres (2018), who identified an association between higher education and better salaries. The higher concentration of residents from the southeastern region observed in the sample may also be related to the high education and income levels, given that Brazil has the highest income concentration (Kawakami et al. 2021).

In the analysis of the results from Tables 3 and 4, an interesting discrepancy is observed between the importance attributed to "adequate nutrition" in the open-ended questions and its average in the quantitative evaluation. In Table 3, the category "Promote adequate nutrition" was the most mentioned, with 143 citations (47.99%), highlighting the perceived importance of adequate nutrition in operational rations by military personnel. However, in Table 4, "nutritionally balanced" and "healthy" had averages of 4.3, not among the attributes with the highest averages.

This difference may suggest a more practical bias among the military personnel. In the responses to the open-ended questions, "Promote adequate nutrition" was frequently cited in association with practical outcomes, such as ensuring operational capability, correcting deficiencies in the current ration, sustaining intense routines, and promoting performance enhancement. For instance, promoting adequate nutrition is mentioned for its intrinsic importance and ability to improve operational efficiency and the health of military personnel in high-energy demand contexts.

Therefore, the analysis of the open-ended question results reveals that mentions of adequate nutrition are often linked to

TABLE 5 | —Comprehensive summary of Nemenyi pairwise comparisons among analyzed variables for product development.

Variable	Sample	Mean and SD	Sum of rankings
Presentation	Bar	3.25 ± 1.80 [a]	1047.00
	Biscuit/cookies	3.51 ± 1.68 [a,b]	1131.00
	Mix grain/cereals	3.51 ± 1.88 [a,b]	1131.00
	Snack	3.82 ± 1.79 [b,c]	1231.00
	Gel Squeeze	4.22 ± 2.14 [c,d]	1359.00
	Cake	4.34 ± 2.08 [d]	1399.00
	Powder	5.33 ± 1.84 [e]	1718.00
Texture	Soft	3.72 ± 1.35 [a]	683.000
	Crunchy	3.40 ± 1.22 [b]	811.000
	Fibrous	2.51 ± 1.36 [c]	1042.00
	Pasty	3.23 ± 1.37 [c,d]	1095.00
	Liquid	2.12 ± 1.08 [d]	1199.00
Flavor	Chocolate	3.22 ± 2.08 [a]	1039.00
	Strawberry	4.05 ± 1.92 [b]	1307.00
	Açai	4.36 ± 2.33 [b,c]	1405.00
	Dulce de leche	4.49 ± 2.22 [b,c]	1446.00
	Citric Fruits	4.54 ± 2.34 [b,c]	1462.00
	Cookies and cream	4.72 ± 2.23 [c]	1522.00
	Vanilla	7.73 ± 2.23 [c]	1524.00
	Other fruits	5.86 ± 2.10 [d]	1887.00
Enrichment	Protein	2.23 ± 1.68 [a]	721.000
	Complex carbohydrate	3.52 ± 2.17 [b]	1136.00
	Vitaminas	4.50 ± 2.12 [c]	1451.00
	Minerals	4.81 ± 2.13 [c]	1549.00
	Fiber	5.09 ± 2.18 [c,d]	1639.00
	Cafein	5.64 ± 2.54 [d,e]	1818.00
	Ômega 3	6.09 ± 2.23 [e]	1964.00
	Maltodextrin	6.10 ± 2.53 [e]	1967.00
	Colagen	6.97 ± 2.1 [f]	2245.00
Source of protein	Whey protein	1.90 ± 1.21 [a]	612.000
	Vegetal protein	3.01 ± 1.27 [b]	832.000
	Casein	2.91 ± 1.02 [b,c]	939.000
	Albumin	2.58 ± 1.09 [c]	970.000
	Other	4.58 ± 0.88 [d]	1477.000

*Different letters in the same column for the same variable indicate significant differences between samples ($p < 0.05$) by the Nemenyi test.

practical and specific objectives, reflecting a concern for performance and operational effectiveness. This utilitarian focus may explain why "nutritionally balanced" and "healthy" did not have the highest averages in the quantitative evaluation despite being highly mentioned categories in the open-ended questions. Military personnel value adequate nutrition primarily to achieve other important operational goals.

Regarding the need to develop a complementary product for operational rations, Brazilian military personnel understand that a balanced ration must meet immediate energy needs while preserving long-term functionality. It includes the correct proportion of macronutrients and micronutrients tailored to the specific physical demands of field missions (Sotelo-Díaz and Blanco-Lizarazo 2019). There is a concern with the nutritional value of the foods included in the ration and their health. In contrast, Anibaldi et al. (2020) found that Australian military personnel do not demonstrate the same concern with health and nutrition, showing inadequate eating habits, especially at the beginning of their careers. It is possible that, daily, both Brazilian and Australian military personnel may be more relaxed about nutrition and health. However, when considering operational rations generally consumed during high-exertion missions, the concern with nutrition may become more evident.

Attention to the transportability and practicality of military foods may be related to their crucial role in operational effectiveness, given the logistical challenges associated with long marches and difficult terrain (Cardello 2003). Light and compact foods that do not compromise mobility are essential. These products must be nutritionally dense, provide essential energy and nutrients, and quickly prepare to minimize exposure to danger. A combatant's efficiency is directly linked to the quality of the nutrition received, with adequate diets enhancing cognitive performance and physical resilience in stressful situations (McClung and Gaffney-Stomberg 2016).

Regarding palatability, De Abreu et al. (2023) reported that Brazilian military personnel tend to reject part of the operational rations offered, often bringing improvised items purchased locally. The flavor and variety of foods can affect caloric intake and morale, a frequently addressed aspect in military literature, which should coincide with the healthiness of the foods, meaning their consumption should not cause adverse effects.

The need for continuous research and development to align military rations with technological advances and best practices in food science and technology is highlighted. It calls for new food technologies and an evidence-based approach to developing food products for the military De Abreu et al. (2024). The analysis corroborates a demand for developing current rations, which have not been reformulated in the national territory since 2008 (Brasil 2008). Concurrently, the results point to critical concerns and preferences of military personnel regarding operational rations. These findings are crucial for product developers, nutritionists, and military decision-makers, aiming for improvements that can directly influence the health and effectiveness of military personnel in the field. It underscores the need for a multidisciplinary approach in ration development, incorporating nutrition, food science, logistics, and psychology knowledge. It is crucial to involve military personnel in product development to ensure that the rations meet their needs and preferences, thus avoiding rejection and under-consumption.

Concerns about food safety and digestibility are frequent among military personnel due to the intense physical and mental demands of their activities. The high-stress environment in which military personnel operate can trigger gastrointestinal disturbances that impact physical health and compromise operational performance and cognitive capacity (Sayers et al. 2021). Choosing foods that are safe and easy to digest is crucial to maintaining the efficiency and well-being of military personnel during prolonged missions and under extreme conditions (Margolis et al. 2014). Adopting operational rations that meet these criteria can help mitigate health risks and ensure that military personnel maintain optimal performance levels.

In military operations, the risk of foodborne illnesses can compromise physical health, cognitive function, and overall troop morale, affecting mission outcomes and generating concerns among the troops. Implementing robust food safety protocols and technologies, such as advanced preservation methods and real-time monitoring systems, is essential to prevent contamination and ensure the nutritional quality of military rations (NATO 2019).

Nutritionally balanced foods are crucial, especially in contexts like the military, where energy and nutrition are essential to maintaining physical and mental performance (Agans et al. 2020). Efficiency in transportation and ease of movement are fundamental, hence the need for compact and lightweight foods. In military operations, the logistics of transporting supplies are crucial. Food products with these characteristics are preferable (J. Knapik et al. 2012). Although palatability is often considered secondary in survival, long-term morale and food acceptability must prevent military personnel from experiencing choice monotony (Lenferna De La Motte et al. 2023).

Despite having the lowest average among the attributes, the low environmental impact still scored between the categories "3—important" and "4—very important," indicating that military personnel know the environmental impact's relevance. However, they consider it less of a priority compared to other attributes. Dominguez et al. (2018) show that burning food waste and military ration packaging contribute to emissions of harmful pollutants, including particulate matter, volatile organic compounds, polycyclic aromatic hydrocarbons, and dioxins/furans. Most emitted particles have a diameter of 2.5 μm, which harms health. Although variations in the composition of rations and their packaging do not significantly alter emissions, replacing specific components can reduce these pollutants. Despite being a lower priority, there is an awareness among military personnel about the importance of environmental impact, which may lead to future initiatives to minimize emissions associated with ration disposal.

Prioritizing the desired characteristics in rations showed that the powder form was the least preferred. It may be related to their lesser stability and durability than other food forms, which is a significant consideration in field conditions. Due to the characteristics of military operations, with high mobility and carrying various materials in the backpacks of combatants, carrying materials with a texture similar to cake can disfigure its primary form, crumbling the product and making it unattractive for consumption. Gel was also less preferred. Despite being highly portable and not requiring additional preparations, it was less preferred than bars, biscuits/cookies, mixed grains/cereals, or snacks, which could be due to their liquid consistency being less satiating than solid foods. The feeling of fullness is not solely dependent on calorie intake but also on the physical properties of the food. Solid foods require more chewing and tend to remain in the stomach longer, promoting greater satiety (Troncoso 2021). While convenient for military personnel who require sustained energy, gels might not provide the prolonged energy release or satiety they need. Gels typically provide a quick release of carbohydrates, which is beneficial for short-term energy needs. Still, they may need more nutritional-density options like mixed grains or cereal bars. Military rations must provide sustained energy and a range of nutrients, including proteins and fats, which might be underrepresented in gel products (Naderi et al. 2023; NATO 2019).

On the other hand, Snacks showed better averages, similar to biscuits/cookies and mixed grains/cereals. Snacks are practical, easy to handle, and boost energy and nutrients quickly. However, compared to other presentations, there is a concern about their nutritional density and long-term sustenance (Arora et al. 2020). Many snacks do not provide the same feeling of satiety as other forms, which can lead to overeating or the need to eat more frequently, impractical in operational situations. Additionally, they can cause glycemic spikes; snacks with high sugar content can cause rapid increases in glucose levels followed by drops, negatively affecting the energy and cognitive function of the combatant (Njike et al. 2016).

Grains/cereals and biscuits/cookies effectively provide quick energy and satisfy hunger between meals. However, functional, practical, and logistical considerations distinguish these options. A mix of grains/cereals and biscuits/cookies share an identical median rank, suggesting comparable acceptance. These foods are typically nutrient-dense and offer both convenience for consumption and comfort, aligning with studies highlighting the importance of food comfort for soldier morale (Maheshwari et al. 2021). While popular, biscuits/cookies possess operational drawbacks. They can break easily, affecting the practicality of consumption and diminishing morale if consumed in a crumbled state.

Moreover, nutritional enrichment per serving can be more challenging due to their composition and manufacturing process, which may need to be more easily adaptable to incorporate certain micronutrients or specific supplements. While nutritious, a mix of grains/cereals might require additional preparation or special packaging to maintain freshness and prevent moisture absorption, which can be impractical in an operational environment. This need for special handling could limit their effectiveness as an on-the-go snack in field conditions.

In contrast, "Bar," the favored option, emerged as a robust choice for operational environments due to its durability, ease of packaging, and resistance to physical deterioration such as squashing or crumbling. They are designed to withstand adverse conditions without losing their shape or texture, which can be critical when soldiers need to rely on the integrity of their food. Furthermore, bars offer portability, nutrient density, and ease of consumption. Tanskanen et al. (2012) observed that adding a nutritious bar item resulted in a lower sensation of hunger after a strenuous training day compared to only rations and improved mood during the latter part of the training. However, the high protein content in the bars may have induced satiety, reducing energy intake from field rations. This finding supports that convenient and energy-efficient foods are valued in operational contexts (De Abreu et al. 2024). Food must be functional, convenient, efficient, durable, and carry high nutrient density. A requirement for ease and rapid consumption in hostile circumstances is mirrored by a preference for food and bars in compact formats, a principle adopted in the elaboration of American bars—First Strike Bar (Stanley et al. 2018).

When presenting food in the military context, it must prioritize functionality and convenience, with clear preferences for compact, ready-to-eat formats, such as squeeze bars and gels. Products that require additional preparation or are less convenient to transport and consume are significantly less favored, highlighting practicality as a key determinant of food acceptance by military personnel in operations.

Texture is a critical sensory quality attribute affecting food perception and acceptance. According to Szczesniak (2002), texture is one of the main parameters of mouthfeel and a key attribute in the pleasure of eating. The texture preference profile is important in developing food products, balancing sensory pleasantness and nutritional/practical requirements, especially for military service (Ruiz-Capillas and Herrero 2021). "Liquid" products scored the highest average, indicating the least preference among the military. The need for leak-proof containers and the possibility of spillage make liquid products less ideal for environments where soldiers are constantly moving (NATO 2019).

Additionally, the logistics of storing and transporting liquids can be a challenge, requiring refrigeration or special storage conditions to maintain stability and food safety, which can be impractical in the field. Liquid products may also add weight and volume to a soldier's backpack, a critical consideration in operations where the load must be minimized (Lenferna De La Motte et al. 2023). The liquid texture might also be perceived as less satiating and inconvenient in all operational conditions, where solid foods are more practical and satisfying (Blundell et al. 2010).

"Pasty" formulations ranked as the second least preferred. They generally require utensils for consumption, which may not be feasible where speed and efficiency are necessary. It can be problematic when military personnel are on the move and do not have time or means to use cutlery (Moody 2020). Pasty foods may become monotonous more quickly than those with varied textures. Dietary monotony can lead to reduced caloric intake and underconsumption, potentially compromising the nutritional status of military personnel (Bolhuis and Forde 2020) (Friedl et al. 2020). For Tepper and Yeomans (2016), attempts to reconcile flavor and sensory response are relevant, as well as the relationship between satiety and the physical presentation of a product; pasty foods may be perceived as less satiating than solid foods, which may then inform preference, especially where satiety is important. Detailed analysis of food texture preferences indicates a clear bias toward soft-textured products.

The "fibrous" presentation did not differ from "pasty." Fibrous foods are recognized for inducing satiety and aiding digestion. However, the fibrous texture can be associated with more difficult-to-digest foods, such as certain raw vegetables, fibrous fruits, fiber-enriched bread, and legumes, such as beans and lentils. Kullen, Iredale, et al. (2016) reported that military personnel generally do not consume large quantities of fiber-containing foods, although it was expected that such individuals would have more rigorous diets. Although beneficial for maintaining satiety during prolonged operations, slower digestion can be perceived as less palatable and more difficult to chew and swallow, negatively impacting eating pleasure and, by extension, military morale (Bolhuis and Forde 2020). There may be little time to eat in combat or intensive training, making easily consumable foods preferable. The fibrous texture can require more chewing time, which is not ideal during periods of high operational demand.

The preference for crunchy foods can be attributed to their association with freshness and quality. Crunchiness is often interpreted as an indicator that the food is in its optimal state for consumption, and this texture can offer immediate gratification. In an operational context, crunchiness can provide a welcome change from the monotony of textures common in long-duration rations, helping to maintain interest and satisfaction with meals. However, there are challenges in incorporating crunchy textures into food products intended for challenging operational environments. The integrity of the crunch must be maintained under various storage and transport conditions, and susceptibility to moisture and other environmental factors must be mitigated through innovations in packaging and preservation technology (Rolls 2005).

"Soft" was the highest-ranked texture. Soft foods are often chosen for their ease of consumption, an important characteristic when military personnel may have limited time to eat and need foods that can be quickly consumed without additional preparation. This convenience can reduce downtime and increase operational efficiency, allowing military personnel to maintain focus on critical tasks with minimal meal interruptions (Pasiakos 2020). It is essential to focus on soft foods that still provide a sense of satiety to avoid those with low satiety. Foods with a pleasing texture can play a significant role in military morale. A tasty and comfortable product can offer a small but significant boost in well-being, especially in an often austere and challenging environment. It can have implications for long-term mental health and immediate operational performance (Malkawi et al. 2018). Therefore, the preference for soft texture can be equated to comfort and convenience in consumption. In a military context, this likely relates to practical issues and ease of consumption on the go. Still, fibrous foods may be assumed to be less palatable or more accessible to consume quickly (Stribiţcaia et al. 2020). Additionally, the soft texture may facilitate digestion and rapid absorption of nutrients, which are essential for maintaining energy levels during long deployments. Soft food products are less likely to cause gastrointestinal discomfort, which can be a risk with harder-to-digest foods under extreme physical stress (Forde and Bolhuis 2022).

The preferences for texture highlighted underscore the importance of aligning the sensory characteristics of food with the operational demands and expectations of military consumers. For the armed forces, it is crucial to consider the nutritional value, sensory acceptability, and convenience of consumption when developing new food products. Future studies should investigate how different textures affect performance and morale in military contexts, using multidisciplinary approaches that integrate sensory science, nutrition, and technical aspects of the profession. As this discussion deepens, future research should focus on understanding how food texture affects the sensory experience, physical and cognitive performance, and psychological well-being of military personnel in operations (Mackenzie-Shalders et al. 2021).

Regarding flavor attributes, "other fruits" flavors may face initial resistance due to hesitation in trying the unknown, accentuated in stressful and demanding environments like military operations where food comfort and familiarity are valued (de Medeiros et al. 2022). The flavors "Vanilla" and "Cookies and Cream" show no significant difference in preference and suggest these flavors could be offered under a single category given their similarity. This lack of distinction could be attributed to the common use of vanilla extract in cream flavors, enhancing sweetness and providing a sense of comfort and familiarity (Lawless and Heymann 2010).

Dulce de leche may be preferred for its intensely sweet flavors, which provide comfort and satisfaction and are crucial in high-tension environments like military operations. Additionally, it has cultural relevance in dairy-producing areas, such as Minas Gerais, Paraná, and Rio Grande do Sul, resonating with regional preferences. Açaí, a fruit typical of Northern Brazil, may be popular due to its unique flavor and nutritional value, particularly antioxidants and energy (Menezes et al. 2011). Citrus flavors can contrast richer flavors in rations, combating monotony and offering vitamin C benefits (Laaksonen et al. 2016).

Strawberry is popular for its balance between sweetness and acidity, offering a refreshing change from intense flavors and being rich in antioxidants that are beneficial for long-term health (Fan et al. 2021). According to Ares et al. (2010), classic flavors like chocolate remain popular due to their pleasurable sensory characteristics, potentially evoking favorable emotional responses. Mielmann et al. (2022) suggest the preference for chocolate stems from its comfort and pleasure. Chocolate's nutrient density provides soldiers comfort and energy in challenging environments (Revelle and Warwick 2009). Dark chocolate, rich in flavonoids, offers antioxidant properties and benefits heart and brain health, enhancing cognition and mood crucial for maintaining mental alertness in strenuous operations (Scholey and Owen 2013).

Regarding nutritional enrichment, lower preferences for collagen may relate to perceptions of benefits or personal preferences. Despite its inclusion in joint health literature, collagen's efficacy remains debated (Jabbari et al. 2022). Low interest in caffeine may stem from its potential adverse effects, like sleep disturbances and anxiety, despite its performance-enhancing benefits (Henning et al. 2011). Although not highly preferred, frequent caffeine consumption is noted among military personnel (Knapik et al. 2022). Maltodextrin and omega-3 require more evaluation for operational rations. Maltodextrin's quick energy release may cause glycemic volatility, which is misaligned with sustained energy demands. Omega-3, while beneficial for cognitive and cardiovascular health, needs careful consideration regarding stability and palatability (Harris 2008). Implementing caffeine, maltodextrin, and omega-3 must ensure benefits without compromising health or functionality.

Fibers, minerals, and vitamins presented intermediate values, reflecting a growing recognition of their importance in preventing deficiencies and promoting health (Yiheng Chen et al. 2018). Micronutrients impact cognitive function and stress resilience, which are crucial for military personnel under intense pressure (NATO 2019). Vitamins, essential for immune functions in stressful environments, are critical components of military rations (Karl et al. 2022). Fibers, crucial for digestive and immune health, highlight the importance of including various nutrients to maintain operational strength (Fu et al. 2022). Complex carbohydrates, necessary for sustained energy, have slightly lower preferences (Murphy et al. 2018). Protein, highly preferred, is recognized as essential for muscle repair and recovery and is crucial for resilience and performance (Pasiakos et al. 2013). Adequate protein intake is vital for muscle mass maintenance, rapid recovery, and immune response, optimizing health and performance (Fernández-Lázaro et al. 2020; Key and Szabo-Reed 2023; McLellan 2013).

Whey protein stands out among protein sources, favored for its high biological value and complete amino acid profile (Phillips et al. 2016). Albumin, known for slower absorption, has a lower preference due to perceived effectiveness and sensory qualities (Falvo 2004). Casein, with slow digestion, is ideal for a constant amino acid supply but less desirable for immediate recovery (Joy et al. 2018; Kim 2020). Plant-based proteins are gaining interest in sustainability, and they face perceptions of inferiority compared to animal-based sources (Hertzler et al. 2020; Liu et al. 2023). As technology advances, plant-based proteins may gain acceptance in operational contexts. The "Other" category, encompassing less conventional protein sources, has the lowest preference, likely due to taste, familiarity, and health perceptions (Onwezen et al. 2021).

Analysis of protein preferences reveals a willingness to consider various options. Whey protein is highly valued for its nutritional qualities, while moderate interest in plant-based proteins suggests openness to alternative approaches. Developing new products should consider bioavailability, efficacy, sustainability, cultural preferences, and impact on well-being and morale.

4.1 | Prototype Modeling

The analysis shows how military preferences for nutritional components are based on personal tastes and a complex interplay of benefit perceptions, physical requirements, and individual preferences. The results reflect a trend toward valuing health, nutrition, and safety in food products, with significant considerations for practicality (weight and volume) and pleasantness (satiety and palatability) in developing new products. In the military context, where efficiency and nutritional adequacy are critical, these conclusions are particularly relevant and imply creating nutritious, tasty, and easy-to-consume foods in challenging environments.

Based on considerations of desirable attributes and characteristics, the conception of a prototype model tailored to the specific demands of military forces has been made possible, providing an in-depth insight into the resulting product. The military's ideal product would be a protein bar with a soft texture and chocolate flavor enriched with proteins, using whey protein as the main protein source.

For example, the information that chocolate is highly preferred must be balanced with whether it can withstand long-term storage and field conditions without spoilage. Combining preference insights with practical criteria will help ensure that the products developed appeal to the military's taste buds and meet their logistical needs. The same thinking should be applied to the other attributes.

Feasibility, economic, and production considerations still need to be evaluated, which suggests the importance of collaborations with companies in the Industrial Defense sector for prototype testing and cost–benefit analyses. This is crucial to ensure that the product meets the military's needs and is economically viable for large-scale production.

The co-creation methodology used in this study facilitated effective collaboration, creating a primary product and guiding the development of secondary alternatives. This collaborative strategy proved extremely valuable, allowing for the exploration of various shapes, flavors, nutritional enhancements, and different protein sources. Directly involving military personnel in the development process ensured that the final products precisely met their specific needs while paving the way for future innovations that could significantly enhance nutrition in operational contexts.

5 | Limitations of the Study

Despite considering that the co-creation methodology successfully produced valuable information about military preferences, as with any research, it is important to consider its limitations. Considerations associated with snowball sampling must be highlighted, even as a viable sampling model for the studied population. Due to its non-probabilistic nature, it is impossible to generalize its results. Furthermore, since individuals share specific characteristics, the variability of possible narratives is limited, and the results may have sampling bias. In this sense, more diverse groups should be used in the data set to overcome this barrier, providing more robustness for making product development decisions.

Additionally, a limitation of the study was the restriction on prototyping the co-created product due to the Brazilian Army's military logistics and security standards. These restrictions compromised extensive testing and adaptation of the prototype to real operational conditions. Despite this, companies responsible for feeding troops must adopt co-creation tools to develop and improve products continuously. Integrating feedback in the co-creation phase can optimize the suitability of products for military needs and improve logistical efficiency. Future research should seek ways to overcome these limitations.

6 | Conclusion

The online co-creation technique effectively generated knowledge about formulating a new product tailored to Brazilian Army personnel's operational demands and consumption profile. The methodology allowed for identifying important attributes related to presentation, flavor, texture, and preferred nutritional enrichment. The combination of structured and unstructured data proved crucial for a better understanding the military's perceptions and allowed for a clearer delineation of the product and the military's needs.

These insights are expected to promote collaborative partnerships between the technological sectors of the Armed Forces and the Defense Industry Base, stimulating innovation and the capacity to develop new products. Moreover, the benefits of this study transcend the military food industry, impacting academic and industrial spheres, promoting interdisciplinarity, and creating an environment conducive to the development of customized food solutions.

Author Contributions

Vitor Luiz F. de Abreu: conceptualization, methodology, formal analysis, writing – original draft preparation, writing – reviewing and editing. **Elson R. Tavares Filho:** data curation, Software, writing – reviewing and editing. **Sabrina S. Monteiro:** visualization, investigation, supervision. **Erick A. Esmerino:** conceptualization, methodology, formal analysis, supervision, project administration, writing – reviewing and editing.

Acknowledgments

We thank the National Council for Scientific and Technological Development (CNPq) and Carlos Chagas Filho Research Support Foundation of the State of Rio de Janeiro (FAPERJ) for the grant provided currently to the first author (Grant 150908/2023-7).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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